

Investigating the Design Space of Affective Touch on the Forearm Area

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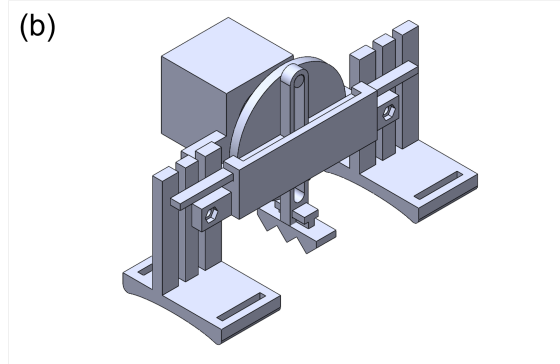
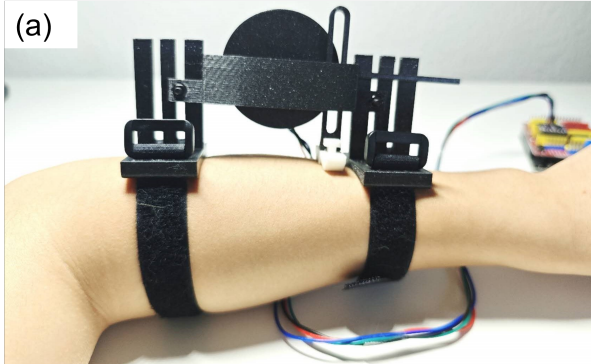


Figure 1: (a). The implementation on a user's forearm of the stroking system; (b). 3D models for the stroking system.

ABSTRACT

Affective touch, which involves slow, gentle mechanical stimulation of the skin, is drawing interest from the Human-Computer Interaction community in recent years. Stroking, the most common form of affective touch, is strongly related to emotional responses and has been proven beneficial for intervening and mitigating anxiety. This has led to a growing need for developing wearable stroking devices. In our study, we first presented a custom-built forearm-worn interface. We then explored the design parameters for stroking devices, focusing on 2 factors: (1) the form factors (shape / material) of the end effector contacted directly with the skin, and (2) the stroking distance, respectively.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; *Empirical studies in HCI*.

KEYWORDS

Haptic device design, Wearable computing, Affective touch, Human-Computer Interaction

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1 INTRODUCTION

Touch represents a fundamental form of communication, with the need for touch persisting throughout an individual's life [4]. The sensory perception of touch enables humans to interact with both the physical environment and other individuals. The design of wearable haptic devices has been extensively explored over the past decades. Historically, researchers have predominantly considered touch to serve a discriminative function. However, empirical evidence has demonstrated that affective touch is strongly associated with emotional experiences [12], such as conveying positive emotions [2, 8, 9] and reducing feelings of isolation [16]. Recent studies have increasingly emphasized the affective value of touch, leading to the development and investigation of numerous wearable devices based on affective touch within the Human-Computer Interaction community [5, 6, 17]. Nevertheless, a significant gap in the current literature exists. While end-effectors like brushes [11] and velvet materials [12] are commonly used in previous works to deliver stroking sensations to users, a direct comparison of end effectors with different shapes or materials is still lacking in wearable computing. In this paper, we address a key research question based on the aforementioned challenge: How do stroking distance and the different types of end-effectors influence the user's perception of the stroking sensation? In this study, we initially developed a

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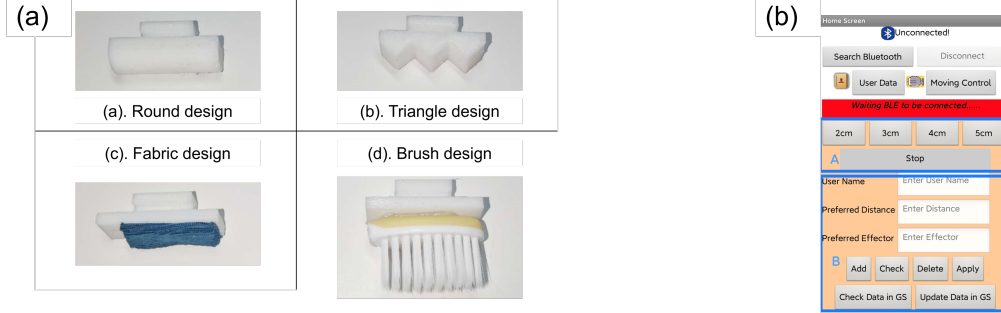


Figure 2: (a). We explore a set of end-effector based on different geometrical parameters and materials; (b) Mobile App interface connected with the smart phone via BLE.

wearable stroking device designed to deliver the stroking sensation to the user's forearm. The stroking distance can be adjusted and customized by the user. Additionally, we designed 4 types of end-effectors, which make direct contact with the user's forearm and feature an easily replaceable design. Subsequently, we conducted an initial user study to gather preliminary insights with 4 participants. We anticipate that the results of our study will offer insights for future research in the design and investigation of wearable stroking devices.

2 IMPLEMENTATION

Stroking Mechanism Design. The stroking device was designed for users on the forearm, as this region is sensitive to tactile stimuli, especially to affective touch [1], and suitable for stable wearable attachments [3, 10, 14]. As demonstrated in Figure 1 (b), the *Scotch Yoke* mechanism is utilized, which converts the rotational motion of a slider into linear motion, and vice versa. The stroking system comprises a slotted yoke and a pin-mounted disc. As the disc rotates, the pin slides within the yoke's slot, generating linear movement. The servo motor (Model: Nema11 Stepper Motor) is directly connected to the disk to drive it to rotate. The housing and moving mechanisms are 3D-printed using PLA. The base of the device, which makes direct contact with the skin, features a curved design to ensure a better fit with individual's forearm. Velcro tape was utilized to mount and secure the device stably to the arm. The linear stroking speed is set at a predetermined rate: 3 ± 0.5 cm/s, chosen to achieve the most pleasant sensation [13]. The ESP32 development board is utilized for motor control and Bluetooth communication with a mobile application Figure 2 (b). For motor driver modules, the Arduino CNC Shield V4 and DRV8825 are selected for the seamless integration with mobile devices.

End Effector Design. We selected two typical end effectors geometries [7]: round and triangle. To deliver affective touch, the most common approach is to use a velvet or soft brush [11, 12, 15]. To keep it as consistent as possible with previous work, we additionally developed two end effectors with varying materials: fabric and soft brush. The design of end effectors are showcased in Figure 2 (a). All the end effectors are 3D printed with soft TPU.

APP Interface Implementation. Two main functions are developed: (1) Control, which allows for customizing the one-way travel distance of the end effector. Pressing the stop button returns the end effector to its initial position immediately and awaits further

commands; (2) Store User Data, which records user information including username, preferred end effector travel distance, and type. The stored data could be sent either via BLE or synchronized with Google Spreadsheets in GS feature.

3 USER STUDY AND RESULTS

Participants and Study design. We recruited 4 participants for the user study (Age range: 26 to 28; Mean age: 27.25, 2 identified as male, 2 identified as female) from the campus through a sample of convenience. None of the participants reported any sensory impairment, and no forearm injuries were present.

Four moving distances are to be tested: 2 cm, 3 cm, 4 cm, and 5 cm, alongside four types of end effector. The testing order is fully randomized for each participant, independent of both end effector type and moving distance. After testing each type of end effector, participants are instructed to rate their experience using a seven-point scale ranging from 1 (very dissatisfied) to 7 (very satisfied) after each moving distance. After experiencing all end effectors, participants were asked to indicate their comments for our device.

Result. Based on the user ratings, the initial results indicate that the round design and fabric design were rated as the most comfortable by participants, followed by the brush design and the triangle design. Regarding the movement distance, 3 cm and 4 cm received the highest ratings, consistent with the findings of the previous study. All participants highlighted that the round design felt more natural and closer to real human touch. One participants reported that "*Due to its sharp points where the skin is in direct contact, the triangle design can cause pain when the stroking distance is increased.*"

4 DISCUSSION AND CONCLUSION

This paper investigates the design space of the wearable stroking device. We hope the initial findings could contribute novel concepts and perspectives to the wearable computing community. For future research, a primary focus will be to augment the sample size of our testing cohorts to bolster the reliability of our outcomes. Furthermore, given the significant correlation between affective touch and emotional states, contextual factors are anticipated to influence our findings. Therefore, we propose further exploration under varied emotional conditions to enhance the robustness and depth of our results. Last but not the least, we aim to enrich the design of end effectors by exploring the interaction of the design factors.

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